



Geographic Information System-Based Assessment of Green Space Accessibility and Urban Livability in Oguta Local Government Area, Nigeria

Temple Probyne Abali^{1*}, Simeon Chukwuemeka Okam¹, Humphery Onuoha², Monday Barine Ainaoba³

¹ Department of Geography and Environment, Rivers State University, 500101 Port Harcourt, Nigeria

² Department of Environmental Management, Rivers State University, 500101 Port Harcourt, Nigeria

³ Institute of Geosciences and Environmental Management, Rivers State University, 500101 Port Harcourt, Nigeria

* Correspondence: Temple Probyne Abali (temple.abali@ust.edu.ng)

Received: 04-18-2026

Revised: 06-01-2026

Accepted: 06-08-2026

Citation: T. P. Abali, S. C. Okam, H. Onuoha, and M. B. Ainaoba, "Geographic information system-based assessment of green space accessibility and urban livability in Oguta Local Government Area, Nigeria," *J. Urban Dev. Manag.*, vol. 5, no. 2, pp. 137–157, 2026. <https://doi.org/10.56578/judm050204>.



© 2026 by the author(s). Published by Acadlore Publishing Services Limited, Hong Kong. This article is available for free download and can be reused and cited, provided that the original published version is credited, under the CC BY 4.0 license.

Abstract: Green space accessibility is an important component of sustainable urban development because of its environmental, health, social, and economic benefits. However, inadequate and uneven access to green spaces remains a challenge in many developing regions, particularly in sub-Saharan Africa. This study assessed green space accessibility in Oguta Local Government Area, Nigeria, and examined its relationship with residents' socio-economic wellbeing and green space utilization. Geographic information system (GIS), remote sensing, and household survey methods were employed. Landsat 8 and Sentinel-2 imagery were used for land use/land cover (LULC) classification and Normalized Difference Vegetation Index (NDVI) analysis, respectively, while buffer and network analyses were conducted to assess green space accessibility. A total of 400 questionnaires were administered, of which 390 were retrieved and 385 were valid for analysis. The results showed that agricultural land (31.8%), forest (26.5%), and wetlands (17.9%) constituted major land-cover classes in the study area. NDVI values ranged from -0.18 to 0.82 , indicating substantial spatial variation in vegetation density. Accessibility analysis showed that 58.7% of respondents were located within the adopted 300 m accessibility threshold, whereas 41.3% were located beyond this distance. Green space accessibility was positively and significantly correlated with respondents' socio-economic wellbeing ($r = 0.641$, $p < 0.001$). Regression analysis also indicated a significant association between green space accessibility and utilization frequency ($p < 0.001$). The findings highlight the importance of improving the spatial accessibility, management, and supporting infrastructure of green spaces to enhance residents' wellbeing and support sustainable urban development in Oguta and environs.

Keywords: Green infrastructure; Urban livability; Geographic information system; Remote sensing; Green space accessibility; Sustainable urban development; Oguta

1 Introduction

Urbanization is one of the most significant demographic and spatial transformations of the twenty-first century. More than half of the world's population currently resides in urban areas, and this proportion is projected to exceed 68% by 2050 [1]. Although urbanization contributes to economic development and infrastructure expansion, it can also exert considerable pressure on vegetation, biodiversity, ecosystem services, environmental quality, and residents' quality of life [2].

Urban green spaces constitute an important component of the urban environment and provide substantial ecological, social, cultural, and economic benefits [3]. They include parks, urban forests, wetlands, gardens, greenways, sports fields, stream corridors, and other vegetated open spaces [4]. These areas contribute to climate regulation, carbon sequestration, biodiversity conservation, habitat provision, air purification, water regulation, and mitigation of the urban heat island effect [5].

Green infrastructure has been recognized as an important component of sustainable urban development because of its contribution to ecosystem services and ecological, social, and economic benefits [6]. Green infrastructure can

also support the development of more sustainable and resilient cities [7]. However, the benefits associated with green spaces depend not only on their availability but also on their accessibility to residents.

Green space accessibility generally refers to the extent to which residents can reach and use available green spaces, particularly within reasonable walking distances from their homes [8]. The World Health Organization has recommended access to public green space within approximately 300 m of residential areas [9]. However, achieving equitable access remains challenging in many rapidly developing urban areas [10]. Accessibility is also closely related to urban livability, which encompasses environmental quality, social wellbeing, access to services and recreational opportunities, and residents' overall quality of life [11, 12]. Previous studies have further demonstrated that accessible green spaces can support recreation, physical activity, social interaction, and individual wellbeing [13].

From an environmental justice perspective, the benefits of urban green infrastructure should be distributed equitably across communities rather than being concentrated in particular locations or population groups. Spatial disparities in green space accessibility may therefore result in unequal opportunities to benefit from environmental amenities and recreational resources. This perspective provides a useful framework for examining differences in green space accessibility among communities and considering their implications for local wellbeing.

Rapid urban growth has placed considerable pressure on natural and vegetated land across many African cities, contributing to reductions and fragmentation of forest and green-space resources [14]. Inadequate urban planning, population growth, infrastructure expansion, and weak environmental management can further reduce the availability and accessibility of urban green spaces [15, 16].

Nigeria has experienced substantial urban and environmental change over recent decades as a result of population growth and urban expansion [17]. Existing studies on urban green infrastructure have largely focused on major Nigerian cities such as Lagos, Abuja, Port Harcourt, and Enugu [18], while comparatively less attention has been given to smaller towns and local government areas. This creates a need for place-specific assessments of green space distribution, accessibility, utilization, and associated socio-economic conditions.

Oguta Local Government Area (LGA) contains diverse natural and semi-natural landscapes that perform important ecological and socio-economic functions. Its forests, wetlands, agricultural areas, riparian vegetation, and the environment surrounding Oguta Lake support agriculture, local biodiversity, recreation, and tourism-related activities [19]. At the same time, increasing human activities, including agricultural expansion, settlement growth, and resource exploitation, may place increasing pressure on these environmental resources [20].

The assessment and sustainable planning of green spaces have therefore received increasing attention in urban and environmental planning [21]. Geographic information system (GIS) and remote sensing provide useful tools for mapping green space distribution, evaluating vegetation condition, and assessing spatial accessibility [22]. When combined with household survey data, geospatial information can also help identify spatial inequalities in accessibility and support evidence-based planning for more equitable green infrastructure provision [22, 23].

Despite the ecological importance of Oguta and its surrounding communities, limited empirical information is available on the spatial accessibility of green spaces, patterns of utilization, and their relationship with residents' perceived socio-economic wellbeing. Therefore, this study evaluates green space distribution and accessibility in Oguta and environs using GIS, remote sensing, and household survey data and examines the relationship between green space accessibility, utilization, and socio-economic wellbeing.

1.1 Research Questions

The study addresses the following research questions:

1. What is the spatial distribution of green spaces in Oguta and environs?
2. What proportion of respondents resides within the adopted walking-distance threshold of a green space?
3. How does green space accessibility vary among the selected communities?
4. How frequently and for what purposes do residents utilize available green spaces?
5. What socio-economic benefits do residents perceive as being associated with accessible green spaces?
6. What constraints affect green space accessibility and utilization in the study area?
7. Is there a significant relationship between green space accessibility and respondents' socio-economic wellbeing?

1.2 Aim and Objectives of the Study

The aim of this study is to evaluate green space accessibility and its socio-economic implications for local communities in Oguta and environs, Imo State, using GIS, remote sensing, and household survey data. The specific objectives are:

1. To assess the spatial distribution of green spaces in Oguta and environs using GIS and remote sensing techniques.
2. To evaluate green space accessibility using buffer and network-based analyses.
3. To examine patterns and frequency of green space utilization among residents.
4. To assess the relationship between green space accessibility and respondents' socio-economic wellbeing.
5. To identify the major constraints affecting equitable access to and utilization of green spaces.

6. To propose strategies for improving sustainable and equitable green space accessibility in the study area.

1.3 Research Hypotheses

The following hypotheses were tested:

Hypothesis 1 (H_1): There is no significant relationship between green space accessibility and respondents' socio-economic wellbeing in Oguta and environs.

Hypothesis 2 (H_2): There is no significant difference in green space accessibility among the selected communities in Oguta and environs.

Hypothesis 3 (H_3): There is no significant association between green space accessibility and the frequency of green space utilization among respondents.

2 Materials and Methods

2.1 Study Area

This study was conducted in Oguta LGA, Imo State, Nigeria. Oguta LGA is located in southeastern Nigeria, at approximately $5^{\circ}40'$ N latitude and $6^{\circ}48'$ E longitude. It covers an area of about 488.0 km^2 [24] and is characterized by an extensive network of rivers, wetlands, forests, agricultural lands, and urban settlements. The area is bounded by Ohaji/Egbema LGA to the north, Ndoni in Rivers State to the south, and neighbouring LGAs within Imo State to the east and west. Figure 1 shows the location of Oguta LGA within Imo State, Nigeria.



Figure 1. Location of Oguta Local Government Area (LGA)

The climate of the area is tropical humid with two distinct seasons: the rainy season (March–October) and the dry season (November–February). Annual rainfall generally exceeds 2,000 mm, while mean annual temperatures range between 26°C and 28°C , creating favourable conditions for dense vegetation growth and wetland development. The study area is drained principally by Oguta Lake, the Njaba River, Orashi River, and several smaller tributaries, all of which contribute to the ecological significance of the region.

Vegetation within Oguta consists predominantly of freshwater swamp forests, riparian vegetation, wetlands, agricultural land, and secondary forests. Urban development, infrastructure development, agricultural activities, and other anthropogenic pressures may affect the distribution and continuity of existing green spaces. These conditions have important implications for environmental sustainability, biodiversity conservation, and residents' accessibility to green infrastructure.

Oguta LGA comprises urban, peri-urban, and rural communities with varying settlement and environmental characteristics. The combination of ecological richness and anthropogenic pressure makes the area particularly suitable for investigating green space accessibility and its implications for urban livability.

2.2 Research Design

The study adopted an integrated geospatial and survey-based research design, combining remote sensing, GIS, and a household questionnaire survey to evaluate green space accessibility and its implications for urban livability in Oguta LGA.

The research design integrated spatial environmental information with socio-economic data collected from residents. This approach enabled the spatial distribution and condition of green spaces to be examined alongside residents' accessibility, utilization patterns, perceived socio-economic benefits, and access-related challenges.

The geospatial component involved the acquisition and processing of satellite imagery to produce land use/land cover (LULC) and Normalized Difference Vegetation Index (NDVI) maps. GIS techniques were subsequently employed to analyse the spatial distribution of green spaces and evaluate accessibility using buffer-based proximity and network-based analyses.

The socio-economic component involved administering structured questionnaires to sampled households across selected communities. The questionnaire collected information on respondents' socio-economic characteristics, green space accessibility, utilization patterns, perceived benefits, and challenges affecting access.

Descriptive and inferential statistical analyses were then conducted to examine patterns in the survey data and assess the relationships among green space accessibility, utilization frequency, and socio-economic wellbeing. The geospatial analyses of LULC and vegetation condition covered the entire Oguta LGA, whereas the household survey and respondent-level accessibility assessment were conducted within the six selected communities. The integration of the geospatial and survey datasets provided a comprehensive framework for examining the spatial and socio-economic dimensions of green space accessibility in the study area. Figure 2 illustrates the overall methodological workflow adopted for the study.

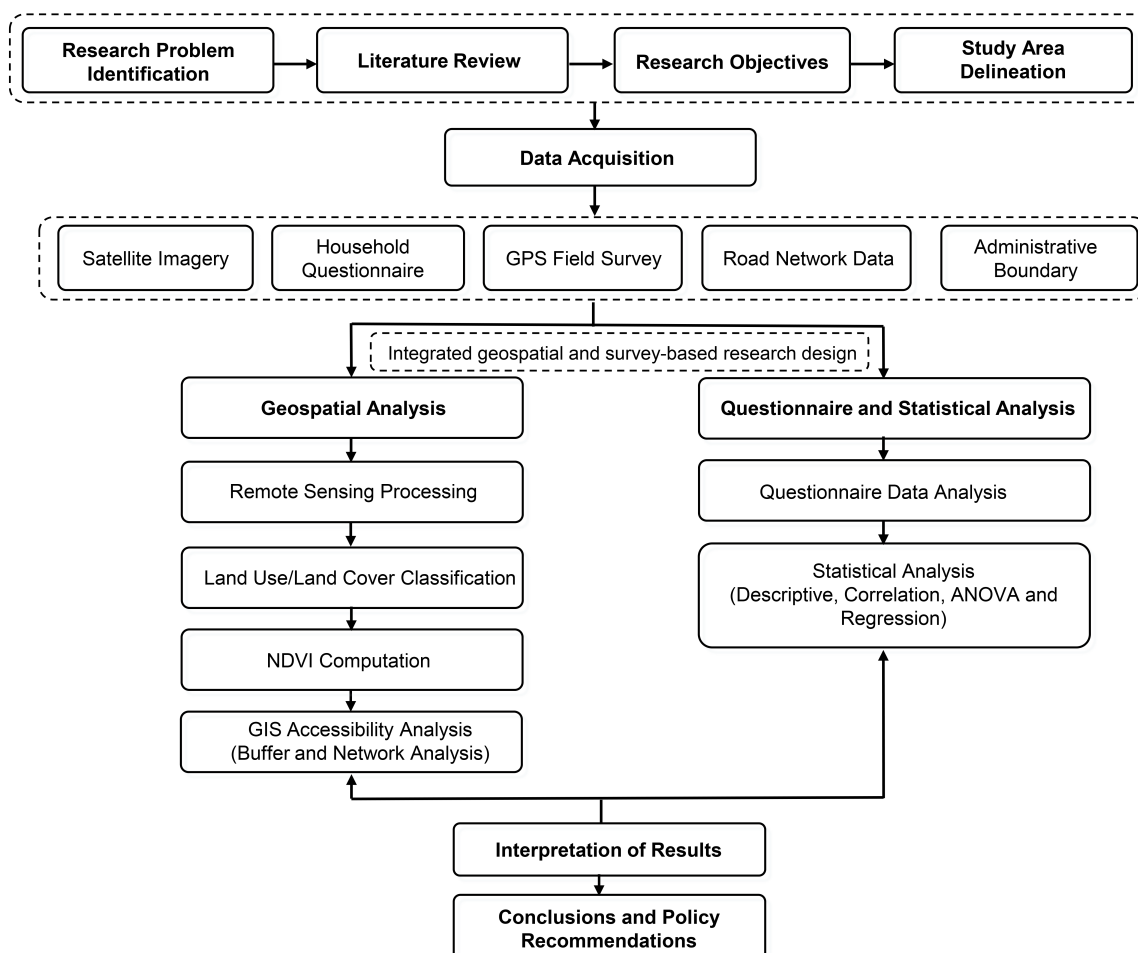


Figure 2. Overall research workflow

Note: GPS = Global Positioning System; GIS = geographic information system; NDVI = Normalized Difference Vegetation Index; ANOVA = analysis of variance.

2.3 Sources of Data

The study utilized both primary and secondary data sources to achieve the research objectives. Primary data were obtained through a structured household questionnaire administered to residents of the selected communities. The questionnaire collected information on respondents' socio-economic characteristics, green space accessibility, frequency and purpose of green space utilization, perceived environmental, health, and socio-economic benefits, and challenges limiting accessibility. Ground-truth observations and Global Positioning System (GPS) coordinates were also collected during field surveys to validate the remotely sensed LULC classification and verify the locations of selected green spaces.

Secondary data included satellite imagery, administrative boundary data, road network data, population statistics, and relevant published literature. Satellite imagery was processed using remote sensing and GIS techniques to produce LULC and NDVI maps, while administrative boundary and road network data were integrated into the GIS environment to support study-area delineation, spatial analysis, and accessibility modelling. A summary of the datasets utilized in the study is presented in Table 1.

Table 1. Summary of data sources

Data	Source	Purpose
Household questionnaire	Field survey	Socio-economic information and accessibility assessment
GPS coordinates	Field survey	Ground truthing and validation
Landsat 8 OLI imagery	USGS EarthExplorer	LULC classification
Sentinel-2 MSI imagery	Copernicus Data Space Ecosystem	NDVI analysis
Administrative boundary map	Government geospatial database	Study area delineation
Road network data	OpenStreetMap	Accessibility and network analysis
Population statistics	National Population Commission (NPC) and community administrative records	Sample design and population estimation

Note: GPS = Global Positioning System; LULC = land use/land cover; MSI = Multispectral Instrument; NDVI = Normalized Difference Vegetation Index; OLI = Operational Land Imager; OSM = OpenStreetMap; USGS = United States Geological Survey.

2.4 Population of the Study

The study was conducted among residents of selected communities within Oguta LGA, Imo State, Nigeria. The total population of Oguta LGA was 142,340 according to the 2006 Population and Housing Census [24] and was projected to approximately 197,900 in 2022 [25]. Eligible respondents for the household survey were adult residents aged 18 years and above because they were considered capable of providing information on green space accessibility, utilization patterns, and the perceived socio-economic benefits associated with green infrastructure.

The sampling frame was developed using population records obtained from the National Population Commission (NPC) and community administrative records. Households served as the primary sampling units. The selected communities represented urban, peri-urban, and rural settlements, thereby capturing variations in the ecological and socio-economic characteristics of the study area.

2.5 Sample Size Determination

For sample-size determination, the sampling population was defined as the combined estimated population of the six selected communities. The estimated population of 159,600 residents was derived from community-level population records and projections obtained from relevant administrative authorities. This population estimate was used as the basis for sample-size calculation because the household survey was conducted only within the six purposively selected communities rather than across the entire Oguta LGA. The sample size for the household questionnaire survey was determined using Yamane's formula for a finite population [26]:

$$n = \frac{N}{1 + N(e)^2} \quad (1)$$

where, n is the required sample size, N is the sampling population (159,600), and e is the acceptable sampling error (0.05).

Substituting the study population into the equation:

$$n = \frac{159600}{1 + 159600(0.05)^2} = \frac{159600}{400} \approx 399$$

The computation produced a minimum sample size of 399 respondents, which was rounded to 400 for questionnaire administration. A total of 400 questionnaires were administered during the field survey. Of these, 390 were retrieved, and 385 were correctly completed and considered valid for analysis, yielding a valid response rate of 96.3%.

2.6 Sampling Procedure

A multistage sampling technique was adopted to select respondents across the selected communities within Oguta LGA.

Stage One: Selection of Communities

In the first stage, six communities were purposively selected based on their proximity to existing green spaces, settlement characteristics, population concentration, and spatial distribution within Oguta LGA. The selected communities represented urban, peri-urban, and rural settlement contexts to capture variations in green space accessibility.

Stage Two: Allocation of Questionnaires

The calculated sample size was proportionately allocated among the selected communities based on their estimated population sizes. The number of questionnaires assigned to each community was determined according to its proportion of the total estimated population. The estimated population figures were used solely for proportional allocation of questionnaires among the selected communities. Table 2 presents the distribution of estimated population, questionnaire allocation, retrieval, and valid responses across the selected communities.

Table 2. Community-wise distribution of questionnaires

Community	Estimated Population	Sample Allocation	Retrieved	Valid
Oguta	26,800	67	66	65
Izombe	25,900	65	64	63
Ezi-Orsu	27,500	69	68	67
Mgbele	24,600	62	61	60
Mgbala Agwa	28,100	70	68	67
Orsu Obodo	26,700	67	63	63
Total	159,600	400	390	385

Source: Authors' computation based on the 2006 National Population Commission (NPC) Population and Housing Census, annual growth-rate projections, and community administrative records obtained from the respective community leadership structures in Oguta LGA.

The population figures presented in Table 2 are estimated community populations rather than official census counts. These estimates were derived from available population records and growth projections and were used solely for proportional allocation of questionnaires among the selected communities. Therefore, they should not be interpreted as official population statistics.

Stage Three: Household Selection

Within each selected community, households were selected using systematic sampling. A sampling interval (I) was determined by dividing the estimated number of eligible households by the allocated sample size for each community. After selecting a random starting point, households were selected at intervals of I until the required sample size for each community was achieved.

Stage Four: Respondent Selection

Where more than one eligible adult resided in a selected household, one respondent was selected using simple random sampling. Eligible respondents were adults aged 18 years and above who had resided in the community for at least one year.

A total of 400 questionnaires were administered, of which 390 were retrieved and 385 were considered valid for analysis. Questionnaires containing substantial missing information or inconsistent responses were excluded from subsequent analyses.

The multistage sampling procedure ensured the inclusion of respondents from communities with different settlement characteristics and levels of green space accessibility.

2.7 Questionnaire Design and Data Collection

Primary data were collected using a structured questionnaire designed to obtain quantitative information on respondents' socio-economic characteristics, green space accessibility, utilization patterns, perceived socio-economic benefits, and constraints affecting accessibility. The questionnaire was developed based on a review of relevant literature on urban green infrastructure, green space accessibility, and urban livability.

The instrument consisted of five sections. Section A collected respondents' demographic and socio-economic information, including gender, age, educational attainment, occupation, and length of residence. Section B examined

green space accessibility, including travel distance, travel time, mode of transportation, and perceived ease of access. Section C assessed the frequency and purpose of green space utilization. Section D evaluated respondents' perceptions of the environmental, social, economic, and health benefits of accessible green spaces, while Section E identified the principal challenges limiting access to green infrastructure.

Most perception-based items were measured using a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). For presentation of the perceived-benefit results, responses 4 and 5 were grouped as Agree, responses 1 and 2 as Disagree, and neutral responses (3) were excluded from the denominator used to calculate the reported percentages. Accessibility was assessed using both GIS-derived spatial indicators and questionnaire-based measures, allowing spatial proximity to be examined alongside respondents' perceived accessibility.

Prior to the main survey, the questionnaire was reviewed by specialists in GIS, urban planning, and environmental management to assess its content validity. A pilot survey was conducted among respondents outside the selected study communities to evaluate the clarity, relevance, and comprehensibility of the questionnaire items. Feedback from the pilot survey was used to revise ambiguous items before the main field survey.

2.7.1 Reliability of the instrument

The internal consistency of the questionnaire constructs was assessed using Cronbach's alpha. A coefficient of 0.70 or higher was considered indicative of acceptable internal consistency. The reliability coefficients for all major questionnaire constructs exceeded this threshold, ranging from 0.79 to 0.87, indicating satisfactory internal consistency (Table 3).

Table 3. Reliability statistics

Construct	Number of Items	Cronbach's Alpha
Accessibility to green spaces	8	0.81
Green space utilization	6	0.84
Socio-economic benefits	10	0.87
Accessibility constraints	6	0.79
Overall questionnaire	30	0.83

Source: Field Survey (2025).

2.8 Remote Sensing Data Acquisition and Processing

Spatial analysis of green infrastructure was conducted using multispectral satellite imagery acquired for the study area. Satellite data were selected based on image quality, spatial resolution, acquisition period, and cloud cover to support LULC classification and vegetation analysis.

Landsat 8 Operational Land Imager (OLI) imagery with a spatial resolution of 30 m was used for LULC classification, while Sentinel-2 Multispectral Instrument (MSI) imagery at 10 m spatial resolution was used for NDVI analysis. The Landsat 8 imagery was acquired on 18 January 2022 from the United States Geological Survey (USGS) EarthExplorer, whereas the Sentinel-2 MSI imagery was acquired on 19 January 2022 from the Copernicus Data Space Ecosystem. The imagery was selected to provide relatively cloud-free coverage of Oguta LGA and to minimize temporal differences between the LULC and vegetation analyses. Images with substantial cloud contamination were excluded, and the selected imagery was clipped to the boundary of Oguta LGA.

The satellite imagery was preprocessed to ensure spatial and radiometric consistency prior to analysis. Preprocessing included radiometric and atmospheric corrections, geometric alignment, and subsetting to the study-area boundary. All spatial datasets were projected to the WGS 1984 / UTM Zone 32N coordinate reference system to ensure spatial consistency and support distance-based analyses. The preprocessing, image classification, and subsequent spatial analyses were performed using ArcGIS Pro, QGIS, and ENVI.

2.9 Land Use/Land Cover Classification

LULC classification was performed using a supervised classification approach based on representative training samples derived from field observations, high-resolution imagery, and existing land use information. Five major LULC classes were identified in the study area: Forest, Wetland, Agricultural land, Built-up area, and Water bodies.

Classification was performed using the Maximum Likelihood Classification (MLC) algorithm. The classifier assigns each pixel to the class with the highest probability based on the spectral characteristics of the training samples. Following classification, post-classification processing was undertaken to reduce isolated pixels and improve the thematic consistency of the final LULC map.

2.9.1 Classification accuracy assessment

The accuracy of the classified LULC map was evaluated using independent reference samples obtained from field observations and high-resolution imagery. A total of 100 independent reference samples were used for accuracy assessment, with 20 samples randomly selected for each LULC class. Classification performance was assessed using overall accuracy, producer's accuracy, user's accuracy, and the Kappa coefficient.

Overall accuracy was calculated as:

$$OA = \frac{N_c}{N} \times 100\% \quad (2)$$

where, OA is the overall accuracy (%), N_c is the number of correctly classified reference samples, and N is the total number of reference samples.

The Kappa coefficient was calculated to assess agreement between the classified map and the reference data beyond chance.

As shown in Table 4, the classification achieved an overall accuracy of 92.0% and a Kappa coefficient of 0.90. Producer's and user's accuracies ranged from 90.0% to 95.0% across the five LULC classes, indicating a high level of agreement between the classified map and the reference data. The resulting LULC map was therefore used in subsequent spatial analyses of green space distribution and accessibility.

Table 4. Confusion matrix for land use/land cover (LULC) classification

Reference Data	Forest	Wetland	Agricultural Land	Built-up Area	Water Bodies	Row Total	Producer's Accuracy (%)
Forest	19	1	0	0	0	20	95.0
Wetland	1	18	1	0	0	20	90.0
Agricultural land	0	1	18	1	0	20	90.0
Built-up area	0	0	1	18	1	20	90.0
Water bodies	0	0	0	1	19	20	95.0
Column total	20	20	20	20	20	100	
User's accuracy (%)	95.0	90.0	90.0	90.0	95.0		

Note: Blank cells indicate values that are not applicable. Source: Field survey and image classification accuracy assessment (2025).

2.10 Normalized Difference Vegetation Index Analysis

Vegetation condition and density were assessed using the NDVI, a widely used spectral index for evaluating vegetation characteristics.

NDVI was calculated as:

$$NDVI = \frac{NIR - Red}{NIR + Red} \quad (3)$$

where, NIR represents near-infrared reflectance and Red represents red-band reflectance.

NDVI values theoretically range from -1 to $+1$. Higher positive values generally indicate denser vegetation, values close to zero are typically associated with sparse vegetation or bare surfaces, while negative values are generally associated with water bodies and other non-vegetated surfaces.

For interpretation, the NDVI values were classified into vegetation-density categories appropriate to the ecological characteristics of the study area. The resulting NDVI map was used to assess the spatial distribution and relative density of vegetation and to complement the LULC analysis.

2.11 Geographic Information System-Based Accessibility Analysis

Green space accessibility was assessed using GIS by integrating green space locations, road network data, residential locations, and administrative boundaries. The analysis was conducted to evaluate the spatial accessibility of existing green spaces within Oguta LGA.

2.11.1 Identification of green spaces

Green spaces were defined as publicly or communally accessible vegetated areas that provide ecological, recreational, aesthetic, or environmental functions. Based on the LULC data, field observations, and available spatial information, the following categories were considered green spaces: urban parks, community parks, forest reserves, riparian vegetation, wetlands with public accessibility, recreational open spaces, and agricultural green areas with public access, where applicable.

Very small vegetated patches and isolated private gardens that were not publicly accessible were excluded from the accessibility analysis.

2.11.2 Geocoding of respondents

The geographic locations of sampled households were recorded using GPS devices during the field survey. The recorded coordinates were imported into the GIS environment and converted into point features representing respondent locations. These locations served as origin points for the accessibility analysis, while the identified green spaces served as destination features.

2.11.3 Road network dataset

Road network data were obtained from OpenStreetMap (OSM) and verified through field observations. The dataset included major roads, secondary roads, local streets, and available pedestrian routes relevant to access to green spaces. Network topology was checked and corrected where necessary to minimize disconnected segments and support route calculations.

2.11.4 Buffer analysis

Three buffer distances were generated around identified green spaces to examine accessibility at different spatial scales: 300 m, representing the principal/convenient walking-access threshold; 500 m, representing a broader walking-accessibility range; and 1 km, representing extended accessibility.

The 300 m threshold was adopted as the principal accessibility criterion for classifying respondents' proximity to green spaces. Respondents located within 300 m of an identified green space were classified as being within the adopted accessibility threshold, whereas those located beyond 300 m were classified as being beyond the threshold. The 500 m and 1 km buffers were used only to provide additional spatial context and were not used to calculate the principal accessibility percentages reported in the results.

2.11.5 Network-based accessibility analysis

Because Euclidean distance does not account for the configuration and connectivity of the road network, network-based accessibility analysis was conducted to provide supplementary information on walking accessibility to green spaces. 5-minute and 10-minute walking service areas were generated around mapped green spaces using the available road network.

These service areas were used to visualize broader network-based accessibility patterns and were treated as supplementary spatial analyses rather than as the basis for respondent-level accessibility classification. The 300 m proximity threshold remained the principal criterion for respondent-level accessibility assessment and classification.

All network-distance and service-area analyses were conducted using the WGS 1984 / UTM Zone 32N projected coordinate reference system. Respondents whose residences were located within 300 m of a green space based on the adopted accessibility criterion were classified as having access within the threshold, while those located beyond 300 m were classified as being beyond the threshold.

The accessibility rate was calculated as:

$$AR = \frac{N_a}{N_t} \times 100\% \quad (4)$$

where, AR is the accessibility rate (%), N_a is the number of respondents within the adopted accessibility threshold, and N_t is the total number of respondents.

The spatial accessibility results were subsequently compared with questionnaire-based measures of perceived accessibility to examine the relationship between objectively measured proximity and residents' perceptions of access.

2.12 Operationalization of Study Variables

To ensure clarity and consistency in the statistical analysis, the principal study variables were operationalized before analysis. Socio-economic wellbeing was treated as a dependent variable and referred to respondents' perceived social and economic benefits associated with access to green spaces. It was measured using composite scores derived from relevant Likert-scale questionnaire items.

Green space utilization was also treated as a dependent variable and was measured according to the frequency with which respondents visited or used green spaces for recreation, exercise, relaxation, tourism, social interaction, or other activities. Utilization frequency was categorized as daily, weekly, monthly, occasionally, or never. For statistical analysis, these categories were coded as 1 = never, 2 = occasionally, 3 = monthly, 4 = weekly, and 5 = daily, with higher scores indicating more frequent green space utilization. Although this variable was originally ordinal, the coded scores were used as a sequential measure of utilization frequency in the regression analysis.

GIS-derived accessibility was represented by the shortest network distance from each respondent's residential location to the nearest mapped green space, expressed in meters, where shorter distances indicated greater spatial accessibility. This continuous distance measure was retained for analyses requiring a quantitative accessibility gradient. In addition, the adopted 300 m accessibility threshold was used to classify respondents into two groups: those residing within 300 m of a green space (coded 1) and those residing beyond 300 m (coded 0). The 300 m classification was

used to represent whether a respondent met the study's defined accessibility threshold rather than as a replacement for the underlying distance measurement.

Perceived accessibility was assessed from questionnaire responses concerning respondents' reported ease of reaching or accessing nearby green spaces. Where multiple questionnaire items were used to measure perceived accessibility, the responses were combined into a perceived-accessibility score by calculating the mean of the applicable Likert-scale items, with higher scores representing greater perceived accessibility. The perceived-accessibility measure was analyzed separately from the GIS-derived measures; no composite accessibility index combining objective and perceived accessibility was created.

Table 5. Operationalization of green space accessibility variables and their use in inferential analyses

Accessibility Measure	Data Source	Measurement/ Operationalization	Coding/Scale	Inferential Analysis	Role in Analysis
GIS-derived network distance	GIS/network analysis	Shortest network distance from respondent's residential location to the nearest mapped green space, measured in meters	Continuous (meters); lower values indicate greater spatial accessibility	Pearson correlation and simple linear regression	Principal objective accessibility predictor; preserves the continuous variation in actual distance
300 m accessibility classification		Respondents were classified according to whether their nearest green space was within the adopted 300 m threshold	1 = within 300 m (accessible); 0 = beyond 300 m (not accessible)	One-way ANOVA for three or more groups; independent-samples t-test for two groups	Threshold-based accessibility classification
Perceived accessibility	Questionnaire	Respondents' reported ease of accessing nearby green spaces based on the relevant questionnaire item(s)	Continuous perceived-accessibility score; higher values indicate greater perceived accessibility	Pearson correlation, regression, or group comparison only where perceived accessibility was the specified variable of interest	Subjective/ perceived accessibility measure
Gender		Respondent's reported sex/gender category	Categorical		
Age		Respondent's age (years) or derived age category	Continuous or categorical, depending on analysis		
Educational attainment		Highest educational level attained	Ordinal/ categorical	Descriptive analysis and relevant group comparisons	Socio-economic characteristic
Occupation		Respondent's principal occupation	Categorical		
Length of residence		Number of years the respondent had lived in the study area	Continuous or categorical		
Household size		Number of persons in the respondent's household	Discrete/ continuous		

Note: GIS = geographic information system; ANOVA = analysis of variance.

The accessibility measure used in each inferential analysis was selected according to the purpose of the statistical test. Pearson correlation analysis used the continuous GIS-derived network distance to the nearest green space to examine the relationship between spatial proximity and the relevant outcome variables. Simple linear regression likewise used GIS-derived network distance in meters as the principal predictor to estimate the association between spatial proximity to green spaces and utilization frequency. Where perceived accessibility was specifically examined as a predictor, the questionnaire-derived perceived-accessibility score was entered as a separate continuous explanatory

variable.

For group-based comparisons, the 300 m accessibility classification was used when comparing respondents within and beyond the adopted threshold. Since this classification produces only two groups, comparisons between these groups were conducted using an independent-samples *t*-test. One-way ANOVA was applied only when three or more accessibility categories were defined.

Socio-economic characteristics, including gender, age, educational attainment, occupation, length of residence, and household size, were collected and examined where appropriate. These variables were used primarily for respondent characterization and, where specified by the research questions, for examining differences in perceived accessibility or other outcome measures. They were not combined with GIS distance or the 300 m classification to form a composite accessibility score. The operational green space accessibility variables and their inferential analyses are summarized in Table 5.

2.13 Statistical Analysis

The collected data were coded, screened, and analyzed using the SPSS, version 29.0. Descriptive statistics, including frequencies, percentages, means, and standard deviations, were used to summarize respondents' socio-economic characteristics, green space accessibility, utilization patterns, and perceived socio-economic benefits. Inferential statistical analyses were subsequently conducted to examine relationships and differences among the study variables.

(1) Pearson Product-Moment Correlation

Pearson correlation analysis was used to assess the relationship between green space accessibility and socio-economic wellbeing. The correlation coefficient was calculated as:

$$r = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum(X_i - \bar{X})^2 \sum(Y_i - \bar{Y})^2}} \quad (5)$$

where, r is the Pearson correlation coefficient, X_i represents the accessibility score, and Y_i represents the socio-economic wellbeing score.

(2) One-Way Analysis of Variance (ANOVA)

One-way ANOVA was used to determine whether green space accessibility differed significantly among the selected communities. Where the overall ANOVA result was statistically significant, Tukey's honestly significant difference (HSD) post hoc test was used to identify significant pairwise differences among communities.

(3) Simple Linear Regression

The coded utilization-frequency score was used as the dependent variable in the regression analysis. Simple linear regression analysis was used to examine the relationship between green space accessibility and green space utilization frequency. Although green space utilization frequency was measured on an ordinal scale, it was coded as a sequential score representing increasing frequency of use. Simple linear regression was applied to estimate the overall association between accessibility and utilization frequency. This analytical approach should be interpreted with caution because the dependent variable is ordinal in nature; therefore, the regression coefficients were interpreted as associations rather than evidence of causality. Future studies may further validate this relationship using ordinal regression models.

The regression model was expressed as:

$$Y = \beta_0 + \beta_1 X + \epsilon \quad (6)$$

where, Y represents green space utilization frequency, X represents the accessibility score, β_0 is the intercept, β_1 is the regression coefficient, and ϵ is the error term.

(4) Assumption Testing

Prior to inferential analysis, relevant statistical assumptions were assessed, including normality, linearity, homoscedasticity, independence of observations, and homogeneity of variance, as applicable to the respective analyses.

All statistical tests were conducted at a 95% confidence level, with statistical significance set at $p < 0.05$. Probability values smaller than 0.001 were reported as $p < 0.001$.

3 Results and Discussion

3.1 Socio-Economic Characteristics of Respondents

A total of 400 questionnaires were administered across the selected communities in Oguta LGA. Of these, 390 were retrieved, and 385 were found to be valid and suitable for analysis, representing a valid response rate of 96.3%. The socio-economic characteristics of the respondents are presented in Table 6.

Table 6. Socio-economic characteristics of respondents

Variable	Category	Frequency	Percentage (%)
Gender	Male	214	55.6
	Female	171	44.4
Age (years)	18–27	82	21.3
	28–37	126	32.7
	38–47	95	24.7
	48–57	53	13.8
	≥58	29	7.5
	Primary	47	12.2
Educational qualification	Secondary	136	35.3
	ND/NCE	79	20.5
	HND/B.Sc.	98	25.5
	Postgraduate	25	6.5

Note: ND = National Diploma; NCE = Nigeria Certificate in Education; HND = Higher National Diploma; B.Sc. = Bachelor of Science.

Male respondents accounted for 55.6% of the sample, while females represented 44.4%, indicating that both genders were substantially represented in the survey. Regarding age, respondents aged 28–37 years constituted the largest group (32.7%), followed by those aged 38–47 years (24.7%) and 18–27 years (21.3%). Respondents aged 48–57 years and those aged 58 years and above accounted for 13.8% and 7.5%, respectively. Overall, the sample was predominantly composed of adults within the younger and middle-age categories.

In terms of educational attainment, 35.3% of respondents had completed secondary education, while 25.5% held a Higher National Diploma or Bachelor of Science (HND/B.Sc.) qualification. Respondents with National Diploma or Nigeria Certificate in Education (ND/NCE) qualifications accounted for 20.5%, those with postgraduate qualifications represented 6.5%, and 12.2% had primary education. Overall, 52.5% of the respondents possessed post-secondary qualifications, indicating a relatively high level of educational attainment among the surveyed population.

3.2 Spatial Distribution of Green Infrastructure

The LULC classification identified five major land-cover categories within the study area: agricultural land, forest, wetlands, built-up areas, and water bodies (Table 7).

Table 7. Land use/land cover (LULC) distribution of Oguta Local Government Area (LGA)

Land Use Class	Area (km ²)	Percentage (%)
Agricultural land	154.4	31.8
Forest	128.6	26.5
Wetlands	86.9	17.9
Built-up area	71.3	14.7
Water bodies	43.8	9.1
Total	485.0	100.0

Agricultural land occupied the largest proportion of the study area (31.8%), followed by forest (26.5%) and wetlands (17.9%). Together, these vegetation-related land-cover classes accounted for more than three-quarters of the mapped landscape, indicating that vegetation-related land covers constitute a substantial proportion of the study area. Built-up areas represented 14.7% of the total land area and were mainly distributed around major settlements and transportation corridors.

The observed land-cover pattern indicates that non-built-up land dominates the study area, with agricultural land and natural vegetation accounting for a substantial proportion of the landscape. However, future expansion of built-up land may progressively fragment existing green infrastructure if urban growth is not guided by effective land-use planning. The predominance of agricultural land, forest, and wetlands highlights the ecological significance of these land-cover types within the study area. Although the study demonstrates substantial vegetation-related land cover at the landscape scale, spatial distribution is equally important. Large areas of vegetation do not necessarily guarantee equitable accessibility for all communities. Consequently, the subsequent accessibility analysis evaluates whether residents are located within acceptable walking distances of available green spaces.

Figure 3 illustrates the spatial distribution of the classified LULC categories across Oguta LGA. The map shows that vegetation-related land cover is widely distributed within and around the Oguta Lake corridor, whereas built-up areas are concentrated around major settlements and the road network. This spatial arrangement partly explains the observed variation in residents' accessibility to green spaces discussed in the following section.

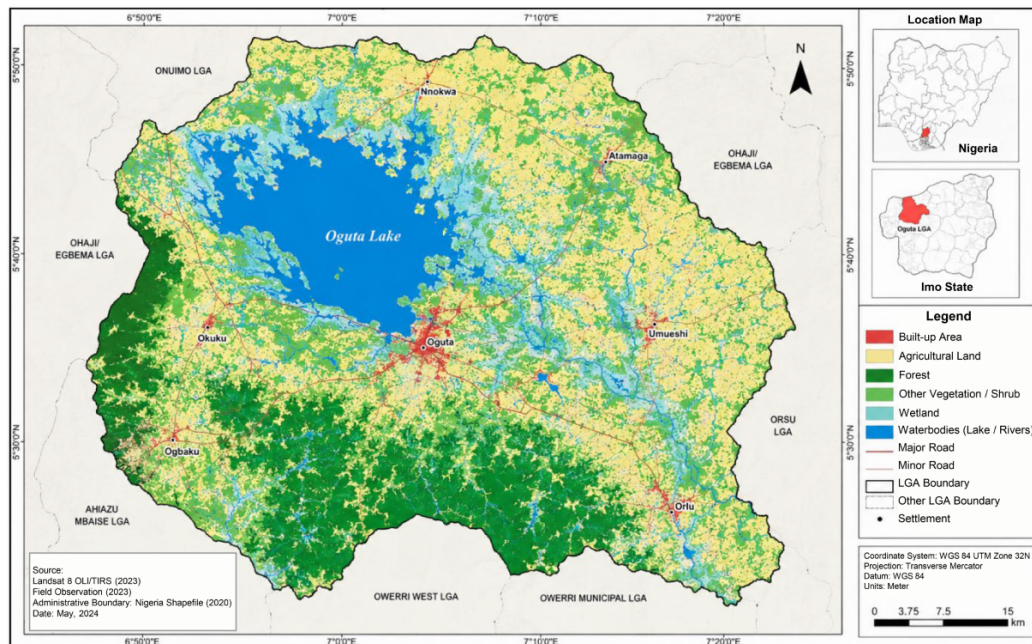


Figure 3. Land use/land cover (LULC) classification map

The predominance of agricultural land, forests, and wetlands demonstrates that Oguta possesses considerable ecological assets capable of supporting environmental sustainability and enhancing urban livability. Nevertheless, future urban development may present potential threats through habitat fragmentation, vegetation loss, and conversion of natural land covers into impervious surfaces.

From an urban planning perspective, these findings highlight the importance of integrating green infrastructure into future development plans to ensure that urban growth does not compromise ecosystem integrity. Protecting existing wetlands, forests, and riparian vegetation will not only conserve biodiversity but also sustain ecosystem services that contribute to climate resilience, flood mitigation, recreational opportunities, and improved public health.

3.3 Green Space Accessibility

Green space accessibility was assessed using the adopted 300 m principal accessibility threshold, supported by additional 500 m and 1 km spatial buffers to examine broader accessibility patterns. Among the 385 valid respondents from the six selected communities, 226 (58.7%) resided within 300 m of an identified green space, whereas 159 (41.3%) resided beyond the 300 m threshold (Table 8). Thus, slightly more than half of the surveyed respondents were within the adopted accessibility threshold, while more than two-fifths were beyond it. The spatial patterns of accessibility based on buffer and network analyses are presented in Figures 4 and 5, respectively.

Table 8. Accessibility to green spaces

Accessibility Category	Frequency	Percentage (%)
Within 300 m	226	58.7
Beyond 300 m	159	41.3
Total	385	100.0

Figure 5 presents the network-based accessibility of green spaces within Oguta LGA using 5-minute and 10-minute walking thresholds. The 5-minute walkable areas are concentrated mainly around the built-up areas and road networks surrounding Oguta Lake, indicating relatively convenient access to nearby green spaces for residents located close to the mapped road network. The 10-minute walkable areas extend beyond the 5-minute zones and cover a wider portion of the study area, demonstrating that accessibility improves as the allowable walking time increases. However, substantial areas, particularly toward the outer and more dispersed settlements, remain outside the 10-minute walkable

zones. This spatial pattern indicates that access to green spaces is unevenly distributed and is strongly influenced by the configuration of the road network and the location of green spaces. Overall, Figure 5 supports the reported finding that a considerable proportion of residents do not have convenient walking access to green infrastructure within the adopted accessibility thresholds.

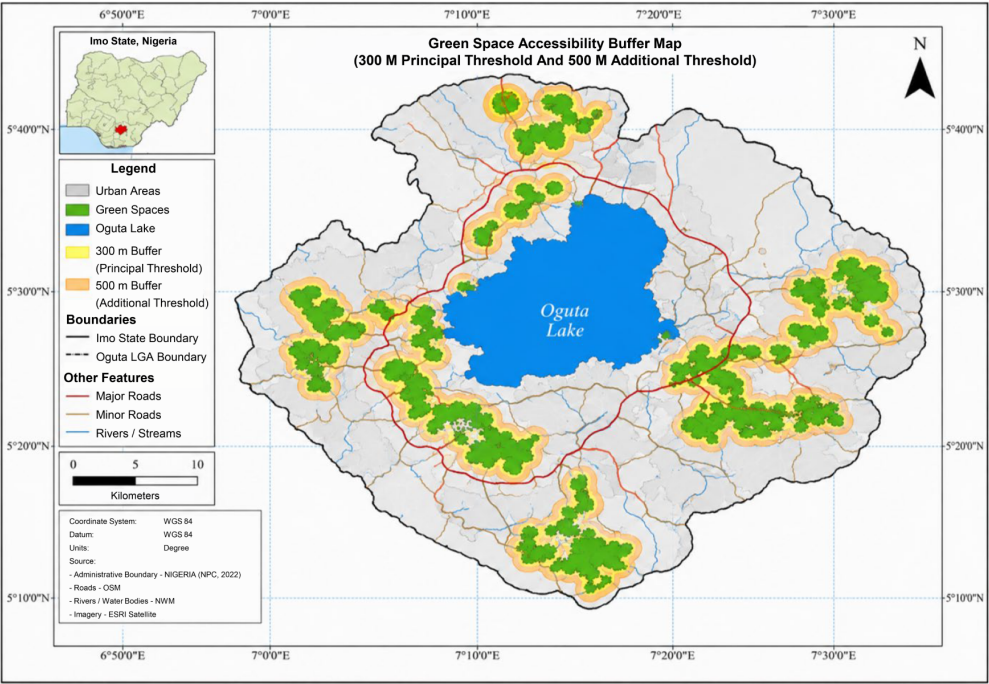


Figure 4. Green space accessibility buffer map showing 300 m and 500 m accessibility zones in OgutaLocal Government Area (LGA)

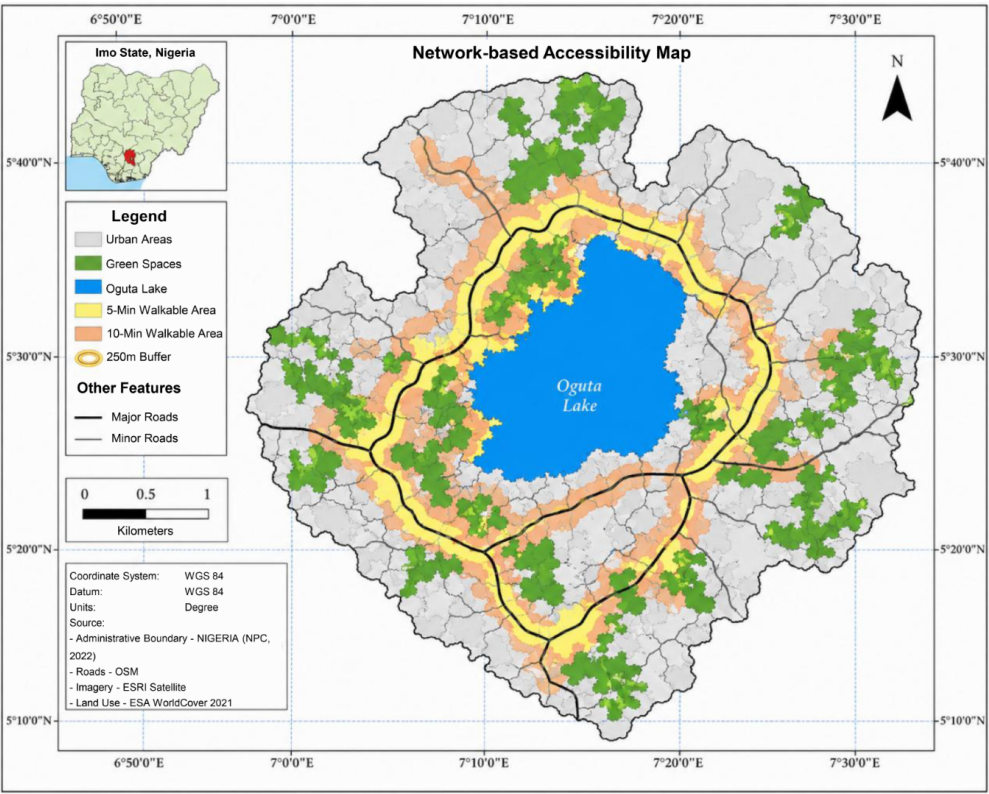


Figure 5. Network-based green space accessibility map of Oguta Local Government Area (LGA)

The results indicate that slightly more than half of the respondents were located within the recommended walking distance of a green space, suggesting relatively good accessibility for a substantial proportion of the study population. Nevertheless, more than two-fifths of respondents were located beyond the 300 m threshold, indicating spatial disparities in the distribution of accessible green infrastructure across the study area.

The accessibility pattern observed is closely related to the uneven distribution of settlements, road infrastructure, and existing green spaces. Communities situated close to Oguta Lake, forest reserves, and other vegetated areas generally enjoyed better physical access than settlements located within more densely built-up areas where accessible green spaces are relatively limited.

Although proximity is an important measure of accessibility, physical distance alone does not necessarily translate into effective access. Respondents identified poor road conditions, inadequate transportation infrastructure, insecurity, and insufficient recreational facilities as additional barriers limiting their use of available green spaces. These findings suggest that accessibility should be viewed as a multidimensional concept involving both spatial proximity and the quality of supporting infrastructure.

The observed spatial disparities in green-space accessibility can be interpreted from an environmental justice perspective, particularly in relation to the equitable distribution of environmental amenities across communities. The substantial proportion of respondents residing beyond the adopted walking-distance threshold suggests that access to the benefits provided by green spaces is not spatially uniform across the study area. Previous studies in Nigeria, South Africa, Europe, and North America have similarly reported that unequal spatial distribution of green spaces may contribute to disparities in recreational opportunities, environmental quality, and perceived quality of life [8, 9, 13, 21, 22]. The findings therefore reinforce the importance of equitable access to green infrastructure in sustainable urban planning. However, accessibility remains uneven across communities, reflecting differences in settlement patterns and the spatial distribution of green spaces and supporting infrastructure.

From a planning perspective, the results suggest that simply increasing the total quantity of green space may not guarantee equitable access. Rather, future urban development should prioritize the strategic distribution of neighbourhood parks and recreational facilities within underserved communities while simultaneously improving transportation infrastructure that connects residents to existing green spaces.

3.4 Vegetation Condition Based on Normalized Difference Vegetation Index Analysis

Vegetation health within the study area was assessed using the NDVI. The NDVI values ranged from -0.18 to 0.82 (Table 9), indicating substantial variation in vegetation density across the landscape (Figure 6).

Table 9. Normalized Difference Vegetation Index (NDVI) vegetation-density classification in Oguta LGA

NDVI Range	Vegetation-Density Category	Area (km ²)	Percentage (%)
$-0.18-0.00$	Non-vegetated / water surface	38.8	8.0
$0.00-0.20$	Sparse vegetation	82.5	17.0
$0.20-0.40$	Moderate vegetation	135.8	28.0
$0.40-0.60$	Dense vegetation	131.0	27.0
$0.60-0.82$	Very dense vegetation	96.9	20.0
	Total	485.0	100.0

Higher NDVI values were recorded within forested areas, wetlands, and riparian vegetation surrounding Oguta Lake, indicating dense and healthy vegetation. Agricultural lands generally exhibited moderate NDVI values within the imagery analyzed. In contrast, built-up areas displayed relatively low NDVI values owing to the predominance of impervious surfaces and limited vegetation cover. Water bodies produced negative or near-zero NDVI values, consistent with the spectral characteristics of open water.

These findings demonstrate that vegetation is unevenly distributed throughout the study area, with the highest vegetation densities concentrated mainly in forested areas, wetlands, and riparian vegetation. Such ecosystems provide important ecological services, including biodiversity conservation, carbon sequestration, erosion control, flood regulation, microclimate moderation, and water-quality protection.

The relatively low NDVI values observed within built-up areas indicate lower vegetation density compared with the more vegetated parts of the study area. Further conversion of vegetated land into impervious surfaces could potentially reduce ecosystem resilience and increase environmental vulnerability.

The NDVI analysis confirms that Oguta retains extensive areas of healthy natural vegetation. The concentration of high NDVI values within wetlands and forest ecosystems highlights the ecological importance of these landscapes and underscores the need for their continued protection. Compared with agricultural and built-up areas, forest and wetland areas generally exhibited higher vegetation density. Consequently, conservation efforts should prioritize these environmentally sensitive areas.

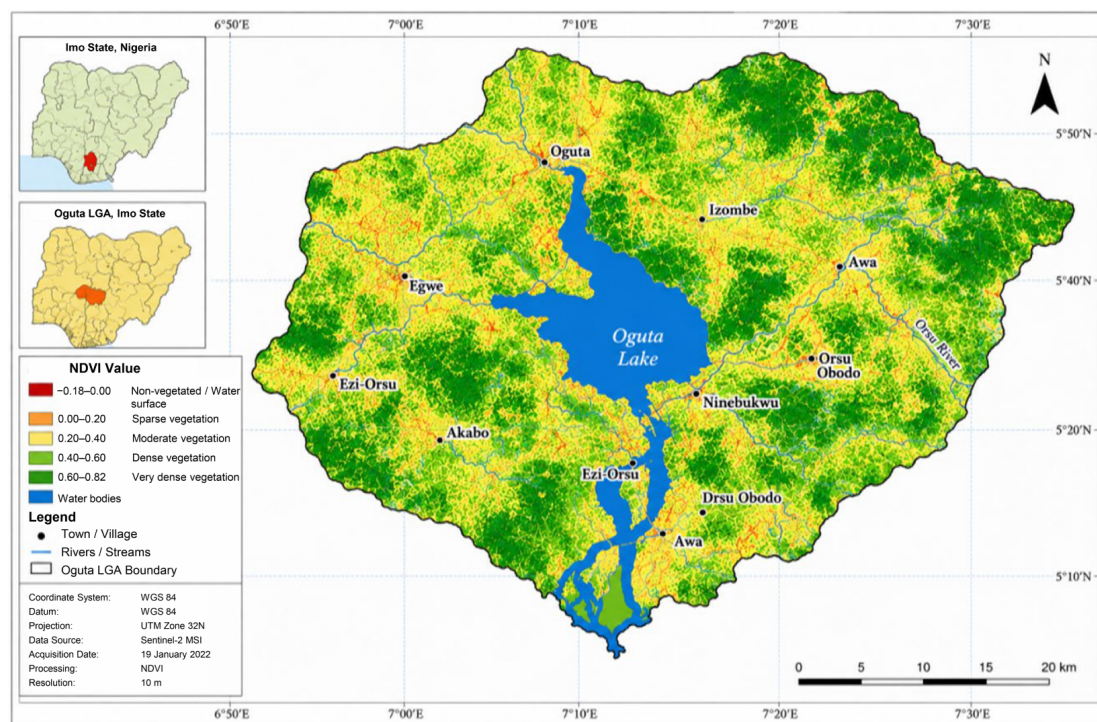


Figure 6. Normalized Difference Vegetation Index (NDVI) map

The results demonstrate the usefulness of remotely sensed vegetation indices for assessing spatial variations in vegetation condition. Overall, the NDVI results complement the land-use analysis by showing not only where vegetation occurs but also the relative condition and density of vegetation across the landscape. Together, these findings provide useful information for urban planners seeking to conserve ecological resources while accommodating future urban growth.

3.5 Green Space Utilization Patterns

The frequency and purpose of green space utilization were analyzed to determine how residents interact with available green infrastructure. The results indicate that green spaces are used primarily for relaxation, exercise, recreation, and social interaction, although the level of utilization varies among respondents (Table 10).

Table 10. Major uses of green spaces

Purpose of Use	Frequency	Percentage (%)
Relaxation	101	26.2
Exercise	84	21.8
Recreation	72	18.7
Social activities	58	15.1
Tourism	42	10.9
Others	28	7.3
Total	385	100.0

Relaxation was the most frequently reported purpose of visiting green spaces (26.2%), followed by exercise (21.8%) and recreation (18.7%). These activities collectively accounted for more than two-thirds of reported green space use, indicating that urban green infrastructure serves primarily as a recreational and wellness resource within the study area. The relatively lower proportion of respondents who visited green spaces for tourism (10.9%) indicates that tourism was a less frequently reported purpose of green-space use compared with relaxation, exercise, and recreation. This pattern may be associated with factors such as recreational facilities, transportation accessibility, or the availability of tourism-related services.

The frequency of visits in Table 11 further illustrates residents' interaction with green spaces. Approximately one-third of respondents (33.5%) reported visiting green spaces weekly, while 17.7% visited daily. Monthly visits

accounted for 26.8%, whereas only 6.2% indicated that they had never visited any green space. These findings demonstrate that green spaces constitute an important component of residents' routine recreational activities.

Table 11. Frequency of green space utilization

Frequency of Visit	Frequency	Percentage (%)
Daily	68	17.7
Weekly	129	33.5
Monthly	103	26.8
Occasionally	61	15.8
Never	24	6.2
Total	385	100.0

The predominance of relaxation and exercise as the primary reasons for visiting green spaces is consistent with contemporary urban planning literature, which recognizes green infrastructure as an essential contributor to physical activity, mental restoration, and social cohesion. Parks and natural environments provide opportunities for stress reduction, leisure, and interpersonal interaction, thereby contributing to improved urban livability. However, utilization patterns should not be interpreted solely as a function of proximity. The frequency with which residents visit green spaces is also influenced by accessibility, safety, availability of recreational facilities, environmental quality, and individual socio-economic characteristics. Consequently, communities located close to green spaces may still exhibit relatively low utilization where supporting infrastructure is inadequate.

The comparatively low proportion of tourism-related visits suggests that tourism is not a predominant purpose of green-space use among the surveyed respondents. Strategic investment in ecotourism infrastructure, environmental conservation, and recreational facilities could improve green-space utilization while supporting sustainable local development.

3.6 Perceived Socio-Economic Benefits of Green Infrastructure

Respondents were asked to assess the socio-economic benefits associated with accessible green spaces. Their responses indicate generally positive perceptions regarding the contribution of green infrastructure to environmental quality, public health, social interaction, and local economic activities, as presented in Table 12.

Table 12. Perceived socio-economic benefits of green infrastructure

Benefit	Agree (%)	Disagree (%)
Improves physical health	84.4	15.6
Reduces stress	81.8	18.2
Enhances environmental quality	88.1	11.9
Promotes social interaction	76.6	23.4
Supports tourism	73.5	26.5
Increases property value	68.6	31.4
Creates employment opportunities	70.1	29.9

The highest level of agreement was observed for environmental improvement (88.1%), followed by improved physical health (84.4%) and stress reduction (81.8%). More than three-quarters of respondents also believed that green spaces promote social interaction, while approximately 70% perceived that they support tourism and employment opportunities. These findings suggest that residents generally recognise the multiple ecosystem services provided by urban green infrastructure. Respondents perceived green spaces not only as recreational environments but also as important environmental assets that contribute to community wellbeing. The high level of agreement regarding environmental and health benefits suggests considerable public awareness of the importance of urban green infrastructure. Such awareness provides a favourable foundation for community participation in conservation programmes and urban greening initiatives.

Although many respondents believed that green spaces contribute to tourism development, employment generation, and increased property values, these responses represent perceived benefits rather than objectively measured economic outcomes. The findings therefore indicate positive resident perceptions of the potential economic contributions of green infrastructure. Overall, the findings are consistent with the multifunctional role of urban green infrastructure in providing environmental, recreational, social, and perceived economic benefits that contribute to urban sustainability.

3.7 Constraints Affecting Green Space Accessibility

Respondents identified several factors limiting effective access to green spaces within the study area. As shown in Table 13, poor road conditions constituted the most frequently identified challenge (24.7%), followed by long travel distance (21.6%) and inadequate maintenance of existing green spaces (19.2%). Security concerns and lack of awareness were reported less frequently but nevertheless represent important barriers to utilization.

Table 13. Challenges affecting green space accessibility

Challenge	Frequency	Percentage (%)
Poor road network	95	24.7
Long distance	83	21.6
Poor maintenance	74	19.2
Inadequate recreational facilities	61	15.8
Security concerns	43	11.2
Lack of awareness	29	7.5
Total	385	100.0

The findings indicate that accessibility extends beyond geographic proximity. Even where green spaces are physically located within recommended walking distances, poor transportation infrastructure, deteriorating facilities, and security concerns may discourage regular visitation. The prominence of road infrastructure as the most frequently reported constraint highlights the importance of integrating transportation planning with green infrastructure development. Investments aimed solely at establishing new parks may yield limited benefits if residents cannot safely and conveniently reach them. Similarly, inadequate maintenance reduces the attractiveness and functionality of existing green spaces. Sustainable urban planning should therefore balance the creation of new green infrastructure with the effective management and maintenance of existing facilities.

Overall, the results demonstrate that improving accessibility requires a comprehensive planning approach encompassing transportation infrastructure, park maintenance, public safety, recreational facilities, and community awareness programmes rather than focusing exclusively on spatial distribution.

3.8 Relationship Between Green Space Accessibility and Socio-Economic Wellbeing

To examine the relationship between green space accessibility and socio-economic wellbeing, Pearson's product-moment correlation analysis was conducted. The results showed a statistically significant positive relationship between the two variables, as presented in Table 14. The analysis revealed a significant positive correlation between green space accessibility and socio-economic wellbeing ($r = 0.641$, $p < 0.001$). This finding indicates that respondents with better access to green spaces generally reported higher levels of socio-economic wellbeing.

Table 14. Pearson correlation between green space accessibility and socio-economic wellbeing

Variables	Pearson's r	p -Value	Decision
Green space accessibility and socio-economic wellbeing	0.641	<0.001	Reject H_1

The observed relationship may reflect the combined influence of several factors, including opportunities for recreation, physical activity, environmental quality, and social interaction that are commonly associated with accessible green spaces. Socio-economic wellbeing may also be influenced by other factors, including income, education, occupation, health status, and neighbourhood characteristics.

The positive association identified in this study is consistent with previous studies reporting relationships between green-space accessibility and residents' perceived quality of life, environmental satisfaction, recreation, and social interaction [5, 8, 13]. Similar associations between proximity to green spaces and physical and psychological wellbeing have also been reported in Europe, North America, and several African cities [9, 14, 15, 17]. Given the cross-sectional design of the study, the findings indicate an association between green-space accessibility and socio-economic wellbeing rather than a causal relationship.

3.9 Hypothesis Testing

H₁: There is no significant relationship between green space accessibility and the socio-economic wellbeing of residents in Oguta LGA.

As shown in Table 14, the Pearson correlation analysis demonstrated a statistically significant positive relationship between green space accessibility and socio-economic wellbeing ($r = 0.641$, $p < 0.001$). Therefore, H_1 was rejected. Respondents with greater green space accessibility generally reported higher levels of socio-economic wellbeing.

H₂: There is no significant difference in green space accessibility among the selected communities in Oguta LGA.

The results of the one-way ANOVA are presented in Table 15. The analysis revealed statistically significant differences in green space accessibility among the selected communities, $F(5, 379) = 5.48$, $p < 0.001$. Therefore, H_2 was rejected. The findings indicate that green space accessibility varies among the selected communities, potentially reflecting differences in settlement patterns, road connectivity, and the spatial distribution of existing green infrastructure.

Table 15. One-way ANOVA for differences in accessibility among communities

Source of Variation	<i>df</i>	<i>F</i>	<i>p</i> -Value
Between communities	5	5.48	<0.001
Within communities	379	—	—
Total	384	—	—

Note: ANOVA = analysis of variance; “—” indicates that the *F*-value and *p*-value are not calculated or reported for the corresponding source of variation.

H₃: There is no significant association between green space accessibility and the frequency of green space utilization.

The regression coefficient and model summary are presented in Tables 16 and 17, respectively. Simple linear regression analysis indicated that green space accessibility significantly predicted utilization frequency, with a positive and statistically significant standardized regression coefficient ($\beta = 0.587$, $t = 14.19$, $p < 0.001$). The model produced a correlation coefficient (*R*) of 0.587, a coefficient of determination (R^2) of 0.345, and an adjusted R^2 of 0.343, indicating that green space accessibility explained approximately 34.5% of the variation in utilization frequency. Therefore, H_3 was rejected. Approximately 65.5% of the variation in utilization frequency was not explained by the model, suggesting that other factors may also contribute to differences in green space utilization. Overall, greater accessibility was associated with more frequent utilization of green spaces.

Table 16. Regression coefficient of green space accessibility predicting utilization frequency

Variable	Standardized β	<i>t</i> -Value	<i>p</i> -Value
Green space accessibility	0.587	14.19	<0.001

Table 17. Regression model summary

<i>R</i>	R^2	Adjusted R^2
0.587	0.345	0.343

Note: *R* = correlation coefficient; R^2 = coefficient of determination.

4 Conclusion

This study assessed green space accessibility and its implications for urban livability in Oguta LGA using GIS, remote sensing, and questionnaire-based survey data. The findings demonstrate that the study area contains substantial vegetation-related land cover, including agricultural land, forests, wetlands, and riparian vegetation. The accessibility analysis indicated that slightly more than half of the respondents resided within the adopted walking-distance threshold of a green space, although accessibility varied among the selected communities. Poor road conditions, long travel distances, inadequate maintenance, and limited recreational facilities were identified as the principal constraints limiting effective access. The statistical analyses revealed significant positive associations between green space accessibility and respondents' perceived socio-economic wellbeing, as well as a significant relationship between accessibility and utilization frequency. These findings indicate that green space accessibility is an important factor associated with both utilization and perceived socio-economic wellbeing. Overall, the study highlights the importance of improving the spatial accessibility, management, and supporting infrastructure of green spaces in promoting sustainable urban development within Oguta LGA.

5 Policy Implications and Recommendations

The recommendations presented below are derived from the empirical findings of this study.

1. **Improve Green Space Accessibility:** Planning authorities should prioritize the provision and strategic distribution of neighbourhood parks and community green spaces, particularly in areas with relatively limited accessibility.
2. **Upgrade Transportation Infrastructure:** Since poor road conditions constituted the most frequently reported barrier to accessibility, improvements in road networks, pedestrian access, and connectivity to existing green spaces should accompany future green infrastructure development.
3. **Strengthen Green Space Management:** Existing parks and recreational facilities should be adequately maintained through regular vegetation management, waste disposal, security provision, and infrastructure rehabilitation to improve user experience.
4. **Protect Ecologically Sensitive Areas:** Wetlands, forests, and riparian vegetation surrounding Oguta Lake should receive enhanced protection because of their ecological significance and relatively high vegetation density.
5. **Integrate GIS and Remote Sensing into Urban Planning:** Planning agencies should employ geospatial technologies to assess green space distribution, monitor future land-use and vegetation changes, evaluate accessibility, and support evidence-based urban planning decisions.
6. **Promote Community Participation:** Local communities should be actively involved in planning, monitoring, and maintaining neighbourhood green spaces to encourage long-term sustainability and stewardship.
7. **Future Research:** Future investigations should incorporate longitudinal study designs, objective health indicators, additional socio-economic variables, more detailed network-based accessibility analyses, and comprehensive remote-sensing accuracy assessments to strengthen understanding of the relationship between green space accessibility and urban livability.

Author Contributions

Conceptualization, T.P.A. and S.C.O.; methodology, T.P.A. and S.C.O.; software, S.C.O.; validation, T.P.A., S.C.O., H.O., and M.B.A.; formal analysis, S.C.O.; investigation, T.P.A., S.C.O., H.O., and M.B.A.; resources, H.O. and M.B.A.; data curation, S.C.O.; writing—original draft preparation, T.P.A. and S.C.O.; writing—review and editing, T.P.A., S.C.O., H.O., and M.B.A.; visualization, S.C.O.; supervision, H.O. and M.B.A.; project administration, T.P.A.; funding acquisition, T.P.A. All authors have read and agreed to the published version of the manuscript.

Informed Consent Statement

Participation in the questionnaire-based survey was voluntary, and respondents were informed about the objectives of the study before data collection. Informed consent was obtained from all participants prior to questionnaire administration.

Data Availability

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request. The satellite imagery used in this study was obtained from the United States Geological Survey (USGS) EarthExplorer and the Copernicus Data Space Ecosystem. Individual-level questionnaire responses and household geolocation data are not publicly available due to participant confidentiality and ethical considerations.

Acknowledgements

The authors sincerely appreciate the residents of Oguta and environs who participated in the survey and provided valuable information for this study. The authors also acknowledge the Department of Geography and Environment, Rivers State University, Port Harcourt, for its institutional and academic support. Appreciation is also extended to the field assistants and community leaders for their assistance during data collection in the study area.

Conflicts of Interest

The authors declare no conflicts of interest.

References

- [1] United Nations, Department of Economic and Social Affairs, “World Urbanization Prospects 2024,” New York, NY, USA: United Nations, 2024. <https://population.un.org/wpp/>
- [2] T. Elmqvist, E. Andersson, N. Frantzeskaki, T. McPhearson, P. Olsson, O. Gaffney, K. Takeuchi, and C. Folke, “Sustainability and resilience for transformation in the urban century,” *Nat. Sustain.*, vol. 2, pp. 267–273, 2019. <https://doi.org/10.1038/s41893-019-0250-1>

- [3] R. Hansen, A. S. Olafsson, A. P. N. van der Jagt, E. Rall, and S. Pauleit, "Planning multifunctional green infrastructure for compact cities: What is the state of practice?" *Ecol. Indic.*, vol. 96, pp. 99–110, 2019. <https://doi.org/10.1016/j.ecolind.2017.09.042>
- [4] N. Kabisch, H. Korn, J. Stadler, and A. Bonn, *Nature-Based Solutions to Climate Change Adaptation in Urban Areas: Linkages Between Science, Policy and Practice*. Cham, Switzerland: Springer, 2017. <https://doi.org/10.1007/978-3-319-56091-5>
- [5] D. Haase, N. Larondelle, E. Andersson, M. Artmann, S. Borgström, J. Breuste, E. Gomez-Baggethun, Å. Gren, Z. Hamstead, R. Hansen *et al.*, "A quantitative review of urban ecosystem service assessments: Concepts, models, and implementation," *Ambio*, vol. 43, pp. 413–433, 2014. <https://doi.org/10.1007/s13280-014-0504-0>
- [6] European Commission, "Green infrastructure (GI)—Enhancing Europe's natural capital," Brussels, Belgium: European Commission, 2013. https://environment.ec.europa.eu/topics/nature-and-biodiversity/green-infrastructure_en
- [7] A. Chiesura, "The role of urban parks for the sustainable city," *Landsc. Urban Plan.*, vol. 68, no. 1, pp. 129–138, 2004. <https://doi.org/10.1016/j.landurbplan.2003.08.003>
- [8] A. Rigolon, "A complex landscape of inequity in access to urban parks: A literature review," *Landsc. Urban Plan.*, vol. 153, pp. 160–169, 2016. <https://doi.org/10.1016/j.landurbplan.2016.05.017>
- [9] J. R. Wolch, J. Byrne, and J. P. Newell, "Urban green space, public health, and environmental justice: The challenge of making cities 'just green enough'," *Landsc. Urban Plan.*, vol. 125, pp. 234–244, 2014. <https://doi.org/10.1016/j.landurbplan.2014.01.017>
- [10] World Health Organization Regional Office for Europe, "Urban green spaces and health," Copenhagen, Denmark: WHO Regional Office for Europe, 2016. <https://www.who.int/europe/publications/i/item/WHO-EURO-2016-3352-43111-60341>
- [11] UN-Habitat, "World cities report 2022: Envisaging the future of cities," Nairobi, Kenya: United Nations Human Settlements Programme, 2022. <https://unhabitat.org/wcr/2022/>
- [12] OECD, *Built Environment Through a Well-being Lens*. Paris, France: OECD Publishing, 2023. <https://doi.org/10.1787/1b5bebf4-en>
- [13] K. Mouratidis, "Urban planning and quality of life: A review of pathways linking the built environment to subjective well-being," *Cities*, vol. 115, p. 103229, 2021. <https://doi.org/10.1016/j.cities.2021.103229>
- [14] D. F. Shanahan, R. Bush, K. J. Gaston, B. B. Lin, J. Dean, E. Barber, and R. A. Fuller, "Health benefits from nature experiences depend on dose," *Sci. Rep.*, vol. 6, p. 28551, 2016. <https://doi.org/10.1038/srep28551>
- [15] C. A. Mensah, "Urban green spaces in Africa: Nature and challenges," *Int. J. Ecosyst.*, vol. 4, no. 1, pp. 1–11, 2014. <https://doi.org/10.5923/j.ije.20140401.01>
- [16] P. B. Cobbinah and R. M. Darkwah, "Urban forest and the question of planning-sustainability inadequacy," *Cities*, vol. 137, p. 104453, 2023. <https://doi.org/10.1016/j.cities.2023.104453>
- [17] E. C. Enoguanbhor, "Geospatial assessments of urban green space protection in Abuja City, Nigeria," *Eximia*, vol. 1, no. 1, pp. 1–15, 2022.
- [18] O. B. Adegun, A. E. Ikudayisi, T. E. Morakinyo, and O. O. Olusoga, "Urban green infrastructure in Nigeria: A review," *Sci. Afr.*, vol. 14, p. e01044, 2021. <https://doi.org/10.1016/j.sciaf.2021.e01044>
- [19] H. M. Ijeomah, O. C. Nwanegbo, and O. Umokoro, "Assessment of tourist attractions in Okomu National Park and Oguta Lake eco-destinations of Nigeria," *PAT*, vol. 11, no. 2, pp. 219–239, 2015.
- [20] G. S. Adaka, A. A. Nlewadim, and J. P. Udoh, "Diversity and distribution of freshwater fishes in Oguta Lake, southeast Nigeria," *Adv. Life Sci. Technol.*, vol. 46, pp. 1–8, 2016.
- [21] C. F. Nwankwo and S. N. Orji, "Spatial assessment of green space accessibility in southeastern Nigeria," *Environ. Monit. Assess.*, vol. 194, no. 6, p. 455, 2022.
- [22] M. O. Abdulraheem, I. O. Oloyede, G. Amuda-Yusuf, W. M. Raheem, A. K. Alade, and M. T. Chukwu, "Urban green space accessibility in Ilorin City, Nigeria," *Int. J. Real Estate Stud.*, vol. 16, no. 1, pp. 24–36, 2022. <https://doi.org/10.11113/intrest.v16n1.108>
- [23] O. J. Ubani, M. O. Alabi, E. N. Chiemelu, A. Okosun, and C. Sam-Amobi, "Influence of spatial accessibility and environmental quality on youths' visit to green open spaces (GOS) in Akure, Nigeria," *Sustainability*, vol. 15, no. 17, p. 13223, 2023. <https://doi.org/10.3390/su151713223>
- [24] National Population Commission, "2006 Population and Housing Census: Priority Table Volume III—Population Distribution by Sex, State, LGA and Senatorial District," Abuja, Nigeria: National Population Commission, 2010. <https://catalog.ihnsn.org/catalog/3340/download/48521>
- [25] Local Governance Accountability Portal, "Oguta LGA, Imo State." <https://lgaportal.org/lgaDetails?lgaId=303>
- [26] T. Yamane, *Statistics: An Introductory Analysis (2nd ed.)*. New York, NY, USA: Harper & Row, 1973.